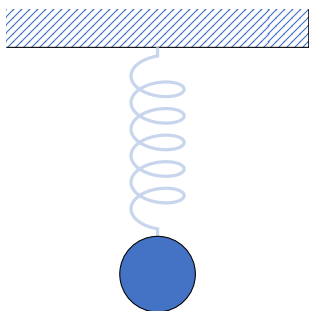


Spring Lab



Purpose We are going to calculate the spring constant k using two different methods. If these two different methods give the same result, we will have confidence that our equations correspond to physical reality.

Ground Rules

- NEVER stretch the spring out with your hands
- NEVER place more than 200 grams on the spring
- When you are not actively taking measurements, do not leave masses hanging on the spring.

1 Finding k using Hooke's Law

In this experiment, we are going to measure how much different masses cause the spring to stretch and use this to determine spring constant.

Procedure

1. Without anything attached to the spring, record its height of the gram
2. Place the 50-gram hanger on the spring and measure height off the ground (after the weight is on the spring). Record in the chart
3. Place different masses on the spring and measure the displacement. *Record in the chart. Make sure all data is in m and kg.*

Data

Total Mass (hanger + mass) (kg)	Force applied to spring ($W = mg$) (N)	Height from the ground (m)
0		
0.050		
0.070		
0.100		
0.130		
0.150		
0.200		

Name: _____

Date: _____

Period: _____

Analysis

Make a graph on Desmos. We are going to plot this a little bit backwards (and you will see why in a moment). **Height should be x , Force should be y .** Plot a best fit line. Write down the equation for each of the best fit line below. You should also copy the link to your graph, paste it on Schoology, and include the link here.

Equation of Line: _____

Link to graph: <https://www.desmos.com/calculator/>_____

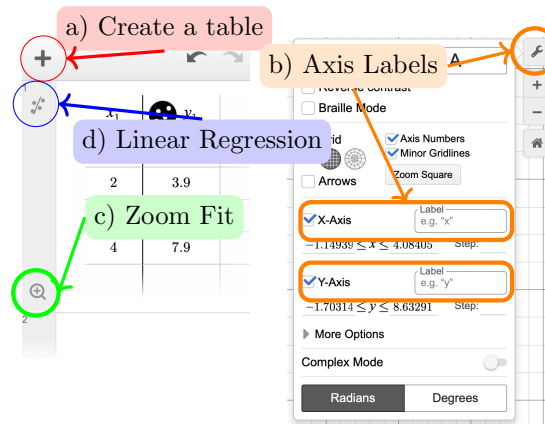
Report your spring constant (with units) and then write a paragraph explaining what you did on this part of the lab. Your paragraph should address: (1) What you graphed and what the graph looked like, (2) why the shape of the graph makes sense, and (3) how you used the graph to find your spring constant.

Spring constant (with units): _____

How to Graph on Desmos

Go to www.desmos.com/calculator to graph your data.

- Start by making a table by clicking the “+” icon at the top left. You will need to create two separate tables.
- Make sure to label the axes using the wrench icon at the right.
- Zoom out so that you can see the whole graph and so that it fills the page. You can do this using the “Zoom Fit” option, but be careful that your fit does not cut off one of the graphs
- Create best fit lines for each graph using the “Linear Regression” tool
- Copy a link to your graph using the export button and clicking “Share a Snapshot”. Write down the link to this graph so you and your instructor can both access your work, later.



Name:

Date:

Period:

2 Finding k using the period of the spring

In this experiment, we are going to measure the spring constant by looking at how the period of the spring varies with mass.

Procedure

1. Pick six different masses. Make sure that they are evenly spread out and take full advantage of the amount of room the spring has to stretch.
2. For each mass, set the spring in motion and measure the time of 5 oscillations. Do this three times and take an average.
3. Calculate the period and square it.

Data

Mass m (kg)	Time of 5 Oscillations (s)				Period T (s)	T^2 (s ²)
	Trial #1	Trial #2	Trial #3	Average		

Analysis

Make another graph on Desmos. This time, **mass should be x and T^2 should be y .** Plot a best fit line. Write down the equation for each of the best fit line below. You should also copy the link to your graph, paste it on Schoology, and include the link here.

Equation of Line:

Link to graph: <https://www.desmos.com/calculator/>_____

Think about what the slope of your line represents. It will be related to the spring constant. To help you, take a look at the rearrangement of the frequency equation below

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (1)$$

$$\frac{1}{f} = T = 2\pi \sqrt{\frac{m}{k}} \quad (2)$$

$$T^2 = \frac{4\pi^2 m}{k} \quad (3)$$

$$T^2 = \left(\frac{4\pi^2}{k} \right) m \quad (4)$$

Name: _____

Date: _____

Period: _____

Report your spring constant (with units) and then write a paragraph explaining what you did on this part of the lab. Your paragraph should address: (1) What you graphed and what the graph looked like, (2) why the shape of the graph makes sense, and (3) how you used the graph to find your spring constant. *For part (3), Don't forget to explain why you did the calculation you did with the slope.*

Spring constant (with units): _____

Conclusion

In theory, both experiments should have yielded the same number for spring constant. We are going to use the percent difference formula to determine how close your two numbers are.

$$\text{Percent Difference} = \frac{|\text{measurement}_1 - \text{measurement}_2|}{\text{avg. measurement}} \times 100\%$$

Calculate the percent difference and then write a paragraph commenting on your percent error. Your paragraph should address: (1) How reasonable is your percent error? (2) Identify three to four factors that could have created uncertainties in your measurements and explain how each factor may have influenced your percent error.

Percent Difference: _____